Chapter 17, Problem 12P

Problem

A voltage source has a periodic waveform defined over its period as

$$v(t) = 120t(2\pi - t)\ \text{V}, \qquad 0 < t < 2\pi$$

Find the Fourier series for this voltage.

Step-by-step solution

Step 1 of 6

The function defined as,

$$v(t) = 120t(2\pi - t)\ \text{V}, \qquad 0 < \text{t} < 2\pi$$

Time period $T = 2\pi$.

Frequency is,

$$\begin{aligned}\omega_0 &= \frac{2\pi}{T}\\ &= \frac{2\pi}{2\pi}\\ &= 1\ \text{rad/sec}\end{aligned}$$

Step 2 of 6

Calculation of a_0.

$$\begin{aligned}a_0 &= \frac{1}{T}\int_0^T f(t)\,dt\\ &= \frac{1}{2\pi}\int_0^{2\pi} v(t)\,dt\\ &= \frac{1}{2\pi}\left[\int_0^{2\pi} 120t(2\pi - t)\,dt\right]\\ &= \frac{120}{2\pi}\left[\int_0^{2\pi} t(2\pi - t)\,dt\right]\\ &= \frac{120}{2\pi}\left[t\left(2\pi t - \frac{t^2}{2}\right)\Bigg|_0^{2\pi} - \int_0^{2\pi}\left(2\pi t - \frac{t^2}{2}\right)dt\right]\\ &= \frac{60}{\pi}\left[2\pi\left(4\pi^2 - \frac{4\pi^2}{2}\right) - 0\left(2\pi 0 - \frac{0^2}{2}\right) - \left(\frac{2\pi t^2}{2} - \frac{t^3}{6}\right)\Bigg|_0^{2\pi}\right]\\ &= \frac{60}{\pi}\left[2\pi\left(2\pi^2\right) - \left(4\pi^3 - \frac{8\pi^3}{6}\right) + (0 - 0)\right]\\ &= \frac{60}{\pi}\left[4\pi^3 - \frac{16\pi^3}{6}\right]\\ &= \frac{60}{\pi}\left(\frac{8\pi^3}{6}\right)\\ &= 80\pi^2\end{aligned}$$

Step 3 of 6

Calculation of a_n.

$$a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t\,dt$$

$$= \frac{2}{2\pi}\int_0^{2\pi} v(t)\cos nt\,dt$$

$$= \frac{1}{\pi}\left[\int_0^{2\pi} v(t)\cos nt\,dt\right]$$

$$= \frac{1}{\pi}\left[\int_0^{2\pi} 120t(2\pi - t)\cos nt\,dt\right]$$

$$= \frac{120}{\pi}\left[\int_0^{2\pi} t(2\pi - t)\cos nt\,dt\right]$$

$$= \frac{120}{\pi}\left[2\pi\int_0^{2\pi} t\cos nt\,dt - \int_0^{2\pi} t^2\cos nt\,dt\right]$$

$$= \frac{120}{\pi}\left[2\pi\left(t\frac{\sin nt}{n}\Big|_0^{2\pi} - \frac{1}{n}\int_0^{2\pi}\sin nt\right) - \left(t^2\frac{\sin nt}{n}\Big|_0^{2\pi} - \frac{2}{n}\int_0^{2\pi} t\sin nt\right)\right]$$

$$= \frac{120}{\pi}\left[\begin{array}{l} 2\pi\left(\frac{2\pi}{n}\sin n2\pi t - \frac{0}{n}\sin n0 - \frac{1}{n}\left(-\frac{\cos nt}{n}\Big|_0^{2\pi}\right)\right) \\ -\left(\frac{4\pi^2}{n}\sin n2\pi - \frac{0}{n}\sin n0 - \frac{2}{n}\left(-t\frac{\cos nt}{n}\Big|_0^{2\pi} + \frac{1}{n}\int_0^{2\pi}\cos nt\right)\right)\end{array}\right]$$

Step 4 of 6

Continue the calculations.

$$= \frac{120}{\pi}\left[\begin{array}{l} 2\pi\left(0 - 0 - \frac{1}{n}\left(\frac{-\cos n2\pi + \cos n0}{n}\right)\right) \\ -\left(0 - 0 - \frac{2}{n}\left(-\frac{2\pi}{n}\cos n2\pi + \frac{0}{n}\cos n0 + \frac{1}{n^2}(\sin n2\pi - \sin n0)\right)\right)\end{array}\right]$$

$$= \frac{120}{\pi}\left[2\pi\left(-\frac{1}{n}\left(\frac{-1+1}{n}\right)\right) - \left(-\frac{2}{n}\left(-\frac{2\pi}{n}(1) + \frac{1}{n^2}(0-0)\right)\right)\right]$$

$$= \frac{120}{\pi}\left[0 - \frac{4\pi}{n^2}\right]$$

$$= \frac{-480}{n^2}$$

Step 5 of 6

Calculation of b_n.

$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega_0 t\,dt$$

$$= \frac{2}{2\pi}\int_0^{2\pi} v(t)\sin nt\,dt$$

$$= \frac{1}{\pi}\left[\int_0^{2\pi} v(t)\sin nt\,dt\right]$$

$$= \frac{1}{\pi}\left[\int_0^{2\pi} 120t(2\pi - t)\sin nt\,dt\right]$$

$$= \frac{120}{\pi}\left[\int_0^{2\pi} t(2\pi - t)\sin nt\,dt\right]$$

$$= \frac{120}{\pi}\left[2\pi\int_0^{2\pi} t\sin nt\,dt - \int_0^{2\pi} t^2\sin nt\,dt\right]$$

$$= \frac{120}{\pi}\left[2\pi\left(-t\frac{\cos nt}{n}\Big|_0^{2\pi} + \frac{1}{n}\int_0^{2\pi}\cos nt\right) - \left(-t^2\frac{\cos nt}{n}\Big|_0^{2\pi} + \frac{2}{n}\int_0^{2\pi} t\cos nt\right)\right]$$

$$= \frac{120}{\pi}\left[\begin{array}{l} 2\pi\left(-2\pi\frac{\cos n2\pi}{n} + 0\frac{\cos n0}{n} + \frac{1}{n}\frac{\sin nt}{n}\Big|_0^{2\pi}\right) \\ -\left(-4\pi^2\frac{\cos n2\pi}{n} + 0\frac{\cos n0}{n} + \frac{2}{n}\left(t\frac{\sin nt}{n}\Big|_0^{2\pi} - \frac{1}{n}\int_0^{2\pi}\sin nt\right)\right)\end{array}\right]$$

$$= \frac{120}{\pi}\left[\begin{array}{l} 2\pi\left(-\frac{2\pi}{n}(1) + 0 + \frac{1}{n^2}(\sin n2\pi - \sin n0)\right) \\ -\left(-\frac{4\pi^2}{n}(1) + 0 + \frac{2}{n}\left(\frac{2\pi}{n}\sin n2\pi - \frac{0}{n}\sin n0 + \frac{1}{n}\frac{\cos nt}{n}\Big|_0^{2\pi}\right)\right)\end{array}\right]$$

Step 6 of 6

Continue the calculations.

$$= \frac{120}{\pi}\left[2\pi\left(-\frac{2\pi}{n} + \frac{1}{n^2}(0)\right) - \left(-\frac{4\pi^2}{n} + \frac{2}{n}\left(0 - 0 + \frac{1}{n^2}(\cos n2\pi - \cos n0)\right)\right)\right]$$

$$= \frac{120}{\pi}\left[-\frac{4\pi^2}{n} - \left(-\frac{4\pi^2}{n} + \frac{2}{n^3}(1-1)\right)\right]$$

$$= \frac{120}{\pi}\left[-\frac{4\pi^2}{n} + \frac{4\pi^2}{n} - 0\right]$$

$$= 0$$

The function $f(t)$ can be expressed as

$$f(t) = a_0 + \sum_{n=1}^{\infty}\left(a_n\cos n\omega_0 t + b_n\sin n\omega_0 t\right)$$

Substitute $80\pi^2$ for a_0, $\frac{-480}{n^2}$ for a_n and 0 for b_n in equation (1).

$$f(t) = 80\pi^2 + \sum_{n=1}^{\infty}\left[\left(\frac{-480}{n^2}\right)\cos nt + 0\sin nt\right]$$

$$= 80\pi^2 - 480\sum_{n=1}^{\infty}\left[\left(\frac{1}{n^2}\right)\cos nt\right]$$

The Fourier series of the voltage is $\boxed{80\pi^2 - 480\sum_{n=1}^{\infty}\left[\left(\frac{1}{n^2}\right)\cos nt\right]}$.